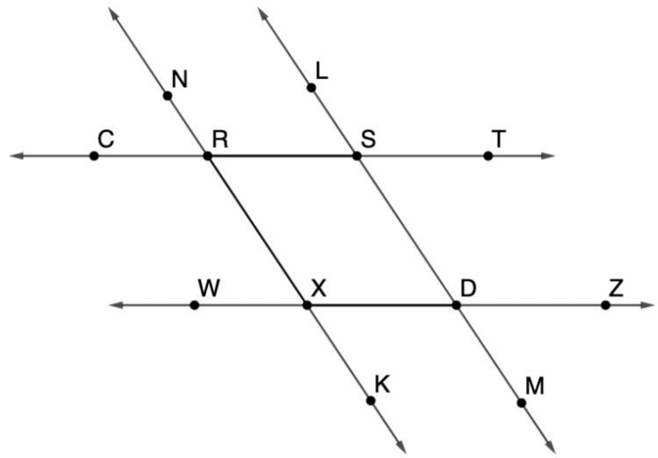
Geometry
Unit 3
Lesson 9 Practice

Name Answer Key________
Date____________________
Period__________________

Complete the two-column proof.



Given: $\overleftrightarrow{NK} || \overleftrightarrow{LM}$ and $\angle LST \cong \angle WXK$

Prove: $\angle CRX \cong \angle RXD$

Statement	Reason
1. $\overleftrightarrow{NK} \| \overleftrightarrow{LM}$	1. Given
2. $\angle WXK \cong \angle XDM$	2. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
3. $\angle LST \cong \angle WXK$	3. Given
4. $\angle LST \cong \angle XDM$	4. Transitive Property of Congruence
5. $\overleftrightarrow{CT} \| \overleftrightarrow{WZ}$	5. If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.
6. $\angle CRX \cong \angle RXD$	6. If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

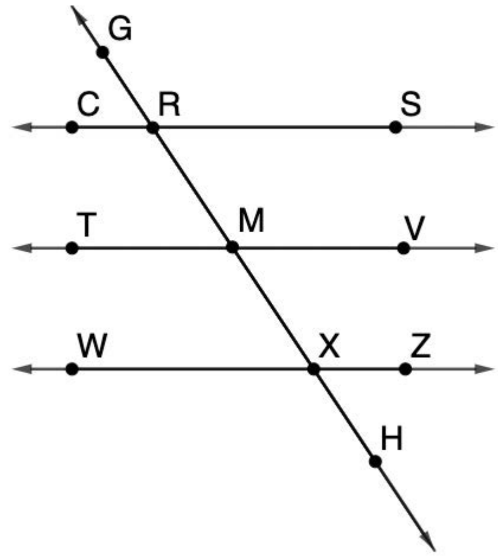
Write the statements and reasons in the correct order in the two-column proof.

$\angle CRX \cong \angle RXD$	Given
$\angle WXK \cong \angle XDM$	Transitive Property of Congruence
$\angle LST \cong \angle XDM$	If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.
$\angle LST \cong \angle WXK$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\overleftrightarrow{CT} \| \overleftrightarrow{WZ}$	If alternate exterior angles are congruent, then the two lines that are intersected by a transversal are parallel.

Complete the two-column proof.

Given: $\angle RMV \cong \angle MXZ$ and
$\angle GRC$ and $\angle WXH$ are supplementary

Prove: $\angle TMX \cong \angle GRS$



Statement	Reason
1. $\angle RMV \cong \angle MXZ$	1. Given
2. $\overleftrightarrow{TV} \parallel \overleftrightarrow{WZ}$	2. If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.
3. $\angle GRC$ and $\angle WXH$ are supplementary	3. Given
4. $\angle MXW$ and $\angle WXH$ are supplementary	4. If two angles form a linear pair, then they are supplementary.
5. $\angle GRC \cong \angle MXW$	5. Supplements of congruent angles are congruent.
6. $\overleftrightarrow{CS} \parallel \overleftrightarrow{WZ}$	6. If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
7. $\angle TMX \cong \angle GRS$	7. If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent

Write the statements and reasons in the correct order in the two-column proof.

$\overleftrightarrow{TV} \parallel \overleftrightarrow{WZ}$	Supplements of congruent angles are congruent.
$\overleftrightarrow{CS} \parallel \overleftrightarrow{WZ}$	If two angles form a linear pair, then they are supplementary.
$\angle TMX \cong \angle GRS$	If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.
$\angle GRC \cong \angle MXW$	If two parallel lines are intersected by a transversal, then the corresponding angles are congruent.
$\angle MXW$ and $\angle WXH$ are supplementary	If corresponding angles are congruent, then the two lines that are intersected by a transversal are parallel.